10TH CLASS COMPUTER SCIENCE — NEW SYLLABUS (PECTAA)

Chapter 2 — System Recovery and Advanced Maintenance

Solved Exercise | New Book (PECTAA)

Exercise (Solved)

A. Multiple Choice Questions

1. Task Manager is mainly used to:

(a) Monitor running applications (b) Clean up disks
(c) Scan for viruses (d) Manage files

Answer: (a) — Task Manager is a built-in tool of Windows which shows real-time information about the running applications and how much CPU, memory and disk they are using. Cleaning the disks and scanning for viruses are the jobs of Disk Cleanup and Windows Defender.

2. Safe Mode starts Windows with:

(a) All drivers and services (b) Basic drivers only
(c) Network services (d) Antivirus tools

Answer: (b) — Safe Mode starts Windows with only the basic drivers and services. It disables the unnecessary drivers and startup software, which makes it easier to fix problems caused by faulty applications or settings.

3. Bootable USB drives are used for:

(a) Installing games (b) System repair and recovery
(c) Watching videos (d) Sending emails

Answer: (b) — A bootable USB drive contains the important system files, so the computer can start from it instead of its own storage drive. It is used for system repair, recovery and reinstallation when Windows cannot start normally.

4. In UEFI systems, a bootable USB must be created in:

(a) NTFS format (b) MBR style
(c) GPT style (d) ZIP format

Answer: (c) — GPT (GUID Partition Table) is the modern partition style, and it is required for UEFI-based systems. MBR is the older style, which is used for older computers.

5. The Power-On Self-Test (POST) is performed:

(a) During system shutdown (b) After login
(c) At system startup (d) When installing software

Answer: (c) — POST is the first thing which the firmware does after the computer is turned on. It checks the most important hardware such as the RAM, keyboard and hard drive, and then the operating system is loaded.

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6. Tenorshare is a tool used to:

(a) Play videos (b) Recover lost files

(c) Create charts (d) Monitor CPU

Answer: (b) — Tenorshare is a third-party data recovery program. It scans the hard drive, USB or memory card to find the files which were recently deleted, and then the user can choose which files to recover.

7. File History is used to:

(a) Track website history (b) Save copies of personal files

(c) Install apps (d) Scan for malware

Answer: (b) — File History is a backup feature of Windows. When it is turned on, the system keeps copies of the user's files at regular intervals on an external drive, so that the files can be recovered even if they are deleted or changed.

8. Windows Defender provides:

(a) File compression (b) Virus protection

(c) Disk formatting (d) Music playback

Answer: (b) — Windows Defender is the built-in security tool of Windows which protects the computer from viruses, malware and other threats. It offers quick scans, full scans and real-time protection.

B. Short Questions

1. Write one key function of the Resource Monitor.

The key function of Resource Monitor is to give a deep and detailed view of how the computer's resources, that is the CPU, memory, disk and network, are being used. It shows exactly which programs and background services are using these resources, so it is very helpful when the computer is running slowly, overheating or using a lot of memory without any clear reason.

2. Define the use of Safe Mode in system recovery.

Safe Mode is a recovery option of Windows which starts the computer with only the basic drivers and services. It disables the unnecessary drivers, extra features and startup software, so a problem caused by a faulty application or a bad driver does not appear. Because of this, Safe Mode is used when the system keeps crashing or restarting, when a program or driver freezes the computer, or when the user has to scan for malware or uninstall a recently installed app.

3. List the steps to create a bootable USB using Rufus.

The steps are the following.

(i) Download the system image or installer from the official website of the operating system provider, for example a Windows ISO or a Linux ISO.

(ii) Download Rufus, which is a free USB tool, from its official site (https://rufus.ie/en/).

(iii) Plug in the USB drive and open Rufus.

(iv) In Rufus, select the plugged-in USB drive and the downloaded ISO file.

(v) Choose **GPT** for modern PCs and **MBR** for older systems, and then click **Start** to create the bootable USB.

4. What happens during the Power-On Self-Test (POST)?

POST (Power-On Self-Test) is the first test which the firmware performs immediately after the computer is switched on. It checks the most important hardware, such as the RAM, the keyboard and the hard drive, to make sure that everything is working properly. If all the parts are fine, the computer continues to load the operating system. If something is wrong, the computer may beep or show an error message on the screen; for example, if the RAM is missing, the computer beeps continuously.

5. Why is the boot order setting important when using recovery tools from a USB drive?

Boot order is the sequence in which the computer looks for a device to load the operating system. Normally the computer tries the internal hard drive first. When we want to recover the system from a bootable USB, the USB drive must be placed at the top of the boot order, otherwise the computer will try to load the damaged Windows from the hard drive and the recovery tools will never start. After changing the order in the Boot tab of the BIOS/UEFI settings, the settings are saved and the computer starts from the USB drive.

6. List the two benefits of running regular disk cleanup.

(i) It removes temporary files, internet files, system cache, thumbnails and the items of the Recycle Bin, so a large amount of storage space is freed.

(ii) Because the drive is no longer full of unnecessary data, the computer works faster and more smoothly, and there is free space for system updates and new programs.

7. How does Windows Defender help in maintaining system health?

Windows Defender is the built-in security tool of Windows which protects the computer from viruses, malware and other threats, so it keeps the system healthy and safe. It offers two types of scans: a **Quick Scan** which checks the common problem areas, and a **Full Scan** which checks the entire system thoroughly. It also provides **real-time protection**, which means that it keeps watching the system in the background and alerts the user if anything suspicious happens.

C. Long Questions

1. Explain the roles of Task Manager and Event Viewer in system maintenance. Give one example for each tool where it can help solve a problem.

Task Manager: Task Manager is a built-in tool of Microsoft Windows which helps the user to monitor the performance of the computer. It shows real-time information about how the processor (CPU), the memory (RAM) and the storage (disk) are being used. The **Processes** tab shows all the active programs and the amount of resources each of them is using, while the **Performance** tab shows detailed graphs for the CPU, memory, disk and network usage. It can be opened by pressing **Ctrl + Shift + Esc**, or by right-clicking the taskbar and selecting Task Manager.

Uses of Task Manager: the user can see how much processing power and memory each program is using, can select a program and click **End Task** to close it when it is not working properly, and can view the programs which are running silently in the background.

Example: if a browser window freezes and the computer becomes slow, Task Manager is opened, the browser is selected in the Processes tab and **End Task** is clicked. The frozen program is closed and the computer becomes normal again.

Event Viewer: Event Viewer is also a built-in tool of Windows which records detailed logs of the activities of the system. It helps the user to check the errors, warnings and important events related to software, hardware and security, with the date and time of each event. It can be opened by pressing **Windows + R**, typing **eventvwr** in the Run dialog box and pressing Enter.

Example: if the computer restarts again and again without any clear reason, the Event Viewer is opened and its logs are checked. The log shows which driver or which program gave an error just before the restart, so the user knows what to repair or remove.

In short, Task Manager shows the **current** condition of the running programs and resources, while Event Viewer keeps the **record of past** events and errors. Both together make the maintenance of the system easy.

2. Your computer is not starting properly due to a system issue. Describe how System Restore or a bootable recovery USB can be used to fix the problem without deleting personal files.

(a) Using System Restore: System Restore allows the computer to go back to an earlier state without deleting the personal files. It works with the help of **restore points**, which are saved versions of the system settings, created before major changes such as a system update or the installation of a driver. If the computer is not starting properly, the following steps are used:

(i) Open the Control Panel, go to **Recovery** and click **Open System Restore** (if Windows does not start, the same option is available from the Advanced Startup / recovery menu).

(ii) A window appears which shows the available restore points. The most recent restore point of the time when the system was working fine is selected.

(iii) Click **Next** and then **Finish**. The computer restarts and restores the selected settings.

Only the system settings, drivers and programs are returned to the earlier condition, while the documents, pictures and other personal files of the user remain safe.

(b) Using a bootable recovery USB: sometimes the computer fails to start completely, and then even Safe Mode and System Restore cannot be opened. In such a case bootable recovery media is used. A bootable USB is a USB flash drive which contains the important system files, and it starts the computer from outside instead of the computer's own storage.

(i) The boot order of the BIOS/UEFI is changed so that the USB drive is tried first, and the computer is started from the USB.

(ii) From the recovery environment, the built-in tools are used, such as **Startup Repair** to fix the problems in loading, **System Restore** to go back with a restore point, the **Command Prompt**, or the **System Image Recovery**.

(iii) If the operating system is badly damaged, the same media is used to reinstall the operating system.

The personal files are not deleted in these steps, because the recovery tools only repair the system files and the settings; however, the user should keep a backup of the important data.

3. Briefly explain any two preventive maintenance techniques that help to keep a computer running smoothly.

(i) Disk Cleanup: after some time the computer collects a large number of unnecessary files such as temporary internet files, system cache, thumbnails and the items of the Recycle Bin. These files take up the storage space and slow down the performance of the system. Disk Cleanup is the built-in tool of Windows which removes such files safely. The tool is opened by typing **cleanmgr** (or Disk Cleanup) in the Windows search bar; then a drive, usually C:, is selected and the boxes of the file types which are to be deleted are checked. For the system files, the option **Clean up system files** is used, which also removes the files left behind by the old Windows updates and by the delivery of updates.

(ii) Installing system and application updates: software updates, also called **patches**, are released regularly by the developers to fix the bugs, add the features and improve the security. There are two types of updates: the operating system updates, for Windows or macOS, and the application updates, for the programs such as the browsers and media players. In Windows the updates are checked by opening **Settings > Update & Security** and clicking **Check for updates**. The available updates are then installed. Keeping the system updated protects it from the viruses and security threats and prevents sudden crashes and failures.

Other preventive tasks are the regular virus scans with Windows Defender, regular backups of the data on an external drive, and cleaning the dust from the hardware so that it does not overheat.

4. Explain how deleted files can be recovered using a third-party software. Also, discuss why recovery may not always be successful.

When a file is deleted, first it is usually found in the Recycle Bin or in the Previous Versions, but sometimes the file is not found there, for example when the Recycle Bin has already been emptied. In such cases a third-party recovery software is used. **Tenorshare** is a free and easy-to-use file recovery program which can scan the hard drive, a USB drive or a memory card to find the files which were recently deleted.

Steps to recover a file with Tenorshare:

(i) Open Tenorshare and select the location from which the file is to be recovered, such as a particular drive or folder.

(ii) Choose the type of the file which is to be recovered, for example pictures or documents.

(iii) Click **Scan** to start searching for the lost files.

(iv) Select the files which are to be recovered and click **Recover** to restore them; they are saved at another safe place.

Why recovery may not always be successful:

(i) If a file was deleted a long time ago, its space may already have been used by other data, and then the file can no longer be recovered.

(ii) Recovery is not always possible if the hard drive is damaged or reformatted, because the record of the file is lost.

(iii) Files may be permanently lost if the Recycle Bin has been emptied and no backup or previous version is available.

That is why the experts advise that no new file should be saved on the same drive after a file has been deleted by mistake, and that regular backups should be kept.

5. A classmate wants to clean up disk space after a large Windows update. Explain how to use Disk Cleanup, including the option to remove system files safely.

Disk Cleanup is the built-in tool of Windows which safely removes the unnecessary files and frees the disk space. After a large Windows update, a big amount of old files remains on the drive, so this tool is very useful. The steps are the following.

(i) Type **cleanup** in the Windows search bar and open the Disk Cleanup tool from the list. A small window opens which asks to choose a drive to clean.

(ii) Select the drive which is to be cleaned, usually **C:**, and click **OK**.

(iii) The tool calculates the space which can be freed, and then a dialog box appears with the list of the file types, such as temporary files, Recycle Bin, downloaded program files and thumbnails. The boxes of the file types which are to be deleted are checked.

(iv) For the system files, the button **Clean up system files** is clicked. The tool scans the drive again and shows the additional items, such as **Windows Update Cleanup**, which contains the files of the previous updates, and the **Previous Windows installation(s)**. These options can be checked safely, because they only remove the old files which are no longer needed by the system.

(v) Click **OK** and then **Delete Files** to remove the selected files. The drive gets free space and the system runs faster.

Note: the option **Clean up system files** only deletes the old system files and the update leftovers; it does not delete the personal documents, pictures or music of the user, so it is a safe way to free a large amount of space.

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